Microeconomics
Course Code: Hum 2101

Lecture 5: Elasticity

Elasticity

Elasticity is a general concept that can be used to quantify the response in one variable when the other variable changes.

The elasticity of A with respect to B $\frac{\% \ \Delta A}{\% \ \Delta B}$

Elasticity is a measure of how much buyers and sellers respond to changes in market conditions.

Elasticity - A measure of the responsiveness of quantity demanded or quantity supplied to a change in one of its determinants

Elasticity

Some purchases, like those for vacation travel, are luxuries that are very sensitive to price changes.

Others, like food or electricity, are necessities for which consumer quantities respond very little to price changes.

The Price Elasticity of Demand

A measure of how much the quantity demanded of a good responds to a change in the price of that good, computed as the percentage change in quantity demanded divided by the percentage change in price.

The **price elasticity of demand** measures how much the quantity demanded responds to a change in price. Demand for a good is said to be *elastic* if the quantity demanded responds substantially to changes in the price. Demand is said to be *inelastic* if the quantity demanded responds only slightly to changes in the price.

Determinants of The Price Elasticity of Demand

- **Availability of Close Substitutes:** Goods with close substitutes tend to have more elastic demand because it is easier for consumers to switch from that good to another. For example, butter and margarine are easily substitutable. A small increase in the price of butter, assuming the price of margarine is held fixed, causes the quantity of butter sold to fall by a large amount. By contrast, because eggs are food without a close substitute, the demand for eggs is less elastic than the demand for butter. A small increase in the price of eggs does not cause a sizable drop in the quantity of eggs sold.
- **Necessities versus Luxuries:** Necessities tend to have inelastic demands, whereas luxuries have elastic demands. When the price of a doctor's visit rises, people will not dramatically reduce the number of times they go to the doctor, although they might go somewhat less often. By contrast, when the price of sailboats rises, the quantity of sailboats demanded falls substantially. The reason is that most people view doctor visits as a necessity and sailboats as a luxury.

Determinants of The Price Elasticity of Demand

- **Definition of the Market:** The elasticity of demand in any market depends on how we draw the boundaries of the market. Narrowly defined markets tend to have more elastic demand than broadly defined markets because it is easier to find close substitutes for narrowly defined goods. For example, food, a broad category, has a fairly inelastic demand because there are no good substitutes for food. Ice cream, a narrower category, has a more elastic demand because it is easy to substitute other desserts for ice cream. Vanilla ice cream, a very narrow category, has a very elastic demand because other flavors of ice cream are almost perfect substitutes for vanilla.
- **Time Horizon:** Goods tend to have more elastic demand over longer time horizons. When the price of gasoline rises, the quantity of gasoline demanded falls only slightly in the first few months. Over time, however, people buy more fuel-efficient cars, switch to public transportation, and move closer to where they work. Within several years, the quantity of gasoline demanded falls more substantially.

Computing the Price Elasticity of Demand

Economists compute the price elasticity of demand as the percentage change in the quantity demanded divided by the percentage change in the price.

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}.$$

For example, suppose that a 10 percent increase in the price of an ice cream cone causes the amount of ice cream you buy to fall by 20 percent. We calculate your elasticity of demand as

$$\text{Price elasticity of demand} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

The Variety of Demand Curves

Economists classify demand curves according to their elasticity.

Demand is considered elastic when the elasticity is greater than 1, which means the quantity moves proportionately more than the price.

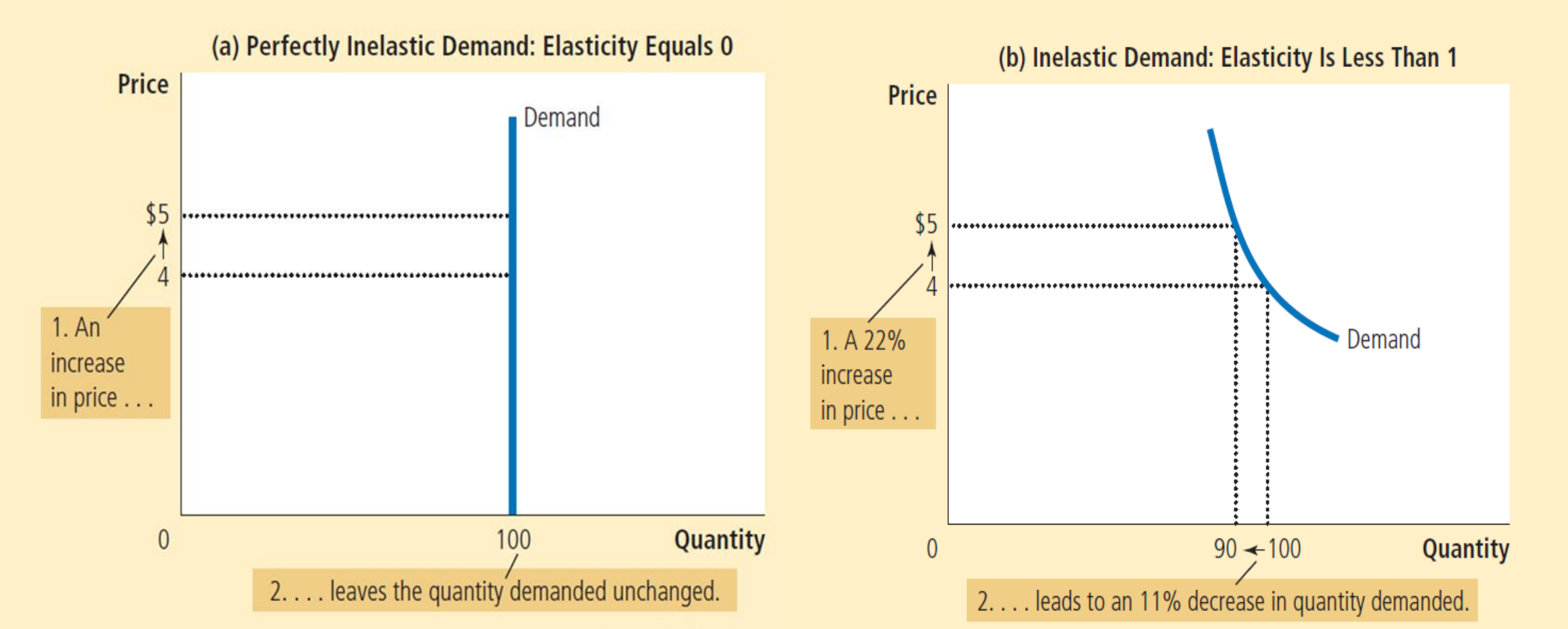
Demand is considered inelastic when the elasticity is less than 1, which means the quantity moves proportionately less than the price.

If the elasticity is exactly 1, the percentage change in quantity equals the percentage change in price, and demand is said to have unit elasticity.

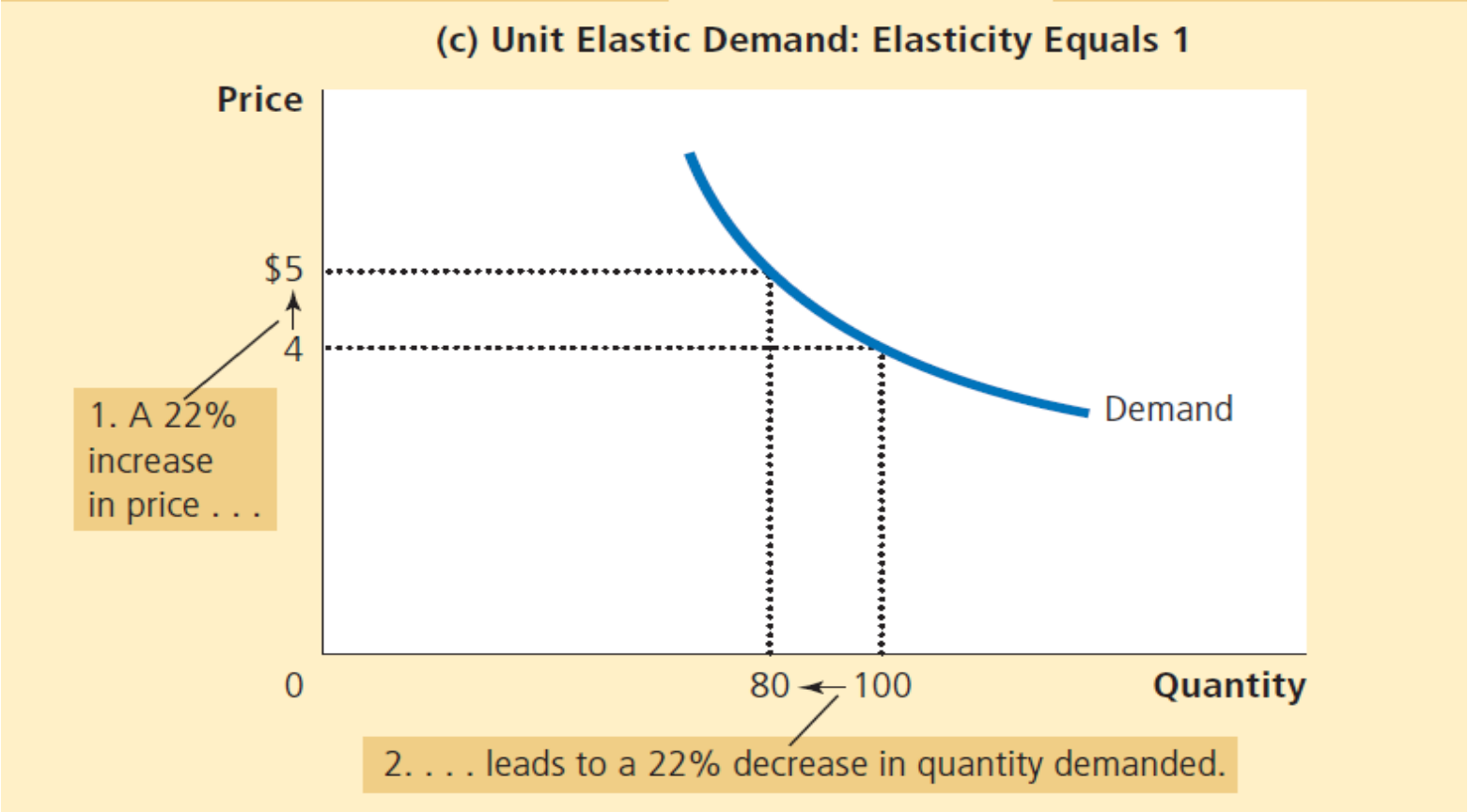
The Variety of Demand Curves

a) Perfectly Inelastic Demand: Elasticity Equals 0
b) Inelastic Demand: Elasticity Is Less Than 1
c) Unit Elastic Demand: Elasticity Equals 1
d) Elastic Demand: Elasticity Is Greater Than 1
e) Perfectly Elastic Demand: Elasticity Equals Infinity

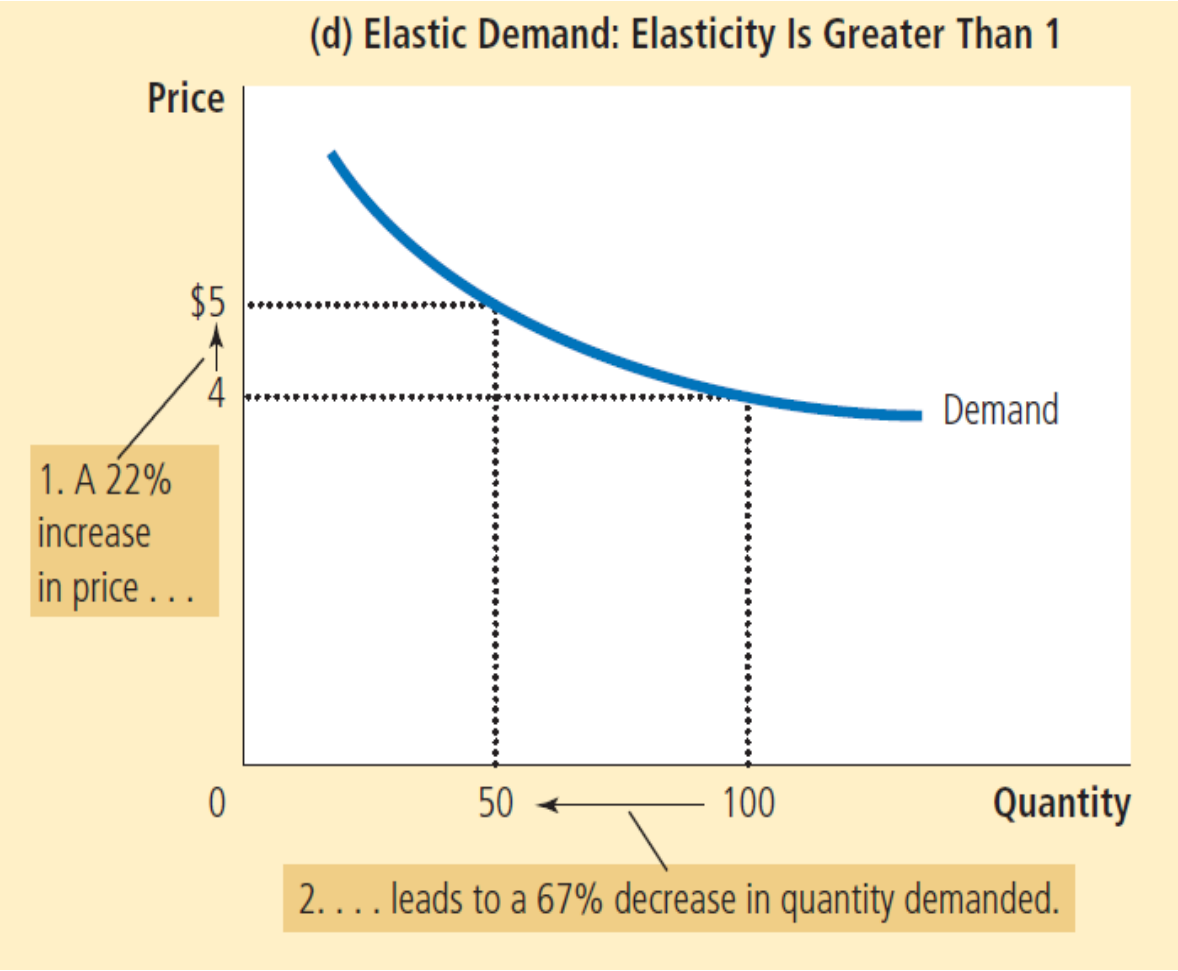
The Variety of Demand Curves

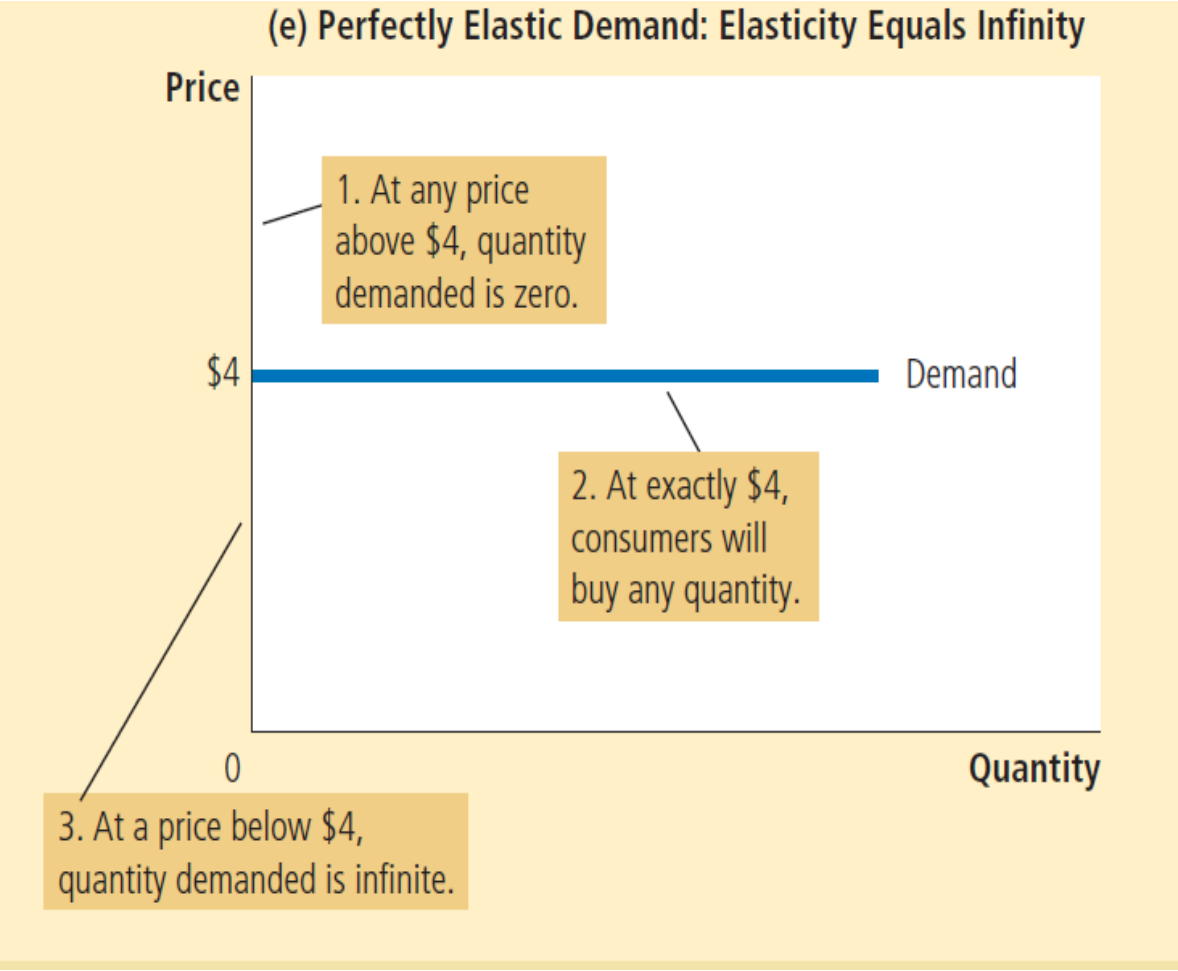


The Variety of Demand Curves



The Variety of Demand Curves





The Price Elasticity of Supply

A measure of how much the quantity supplied of a good responds to a change in the price of that good, computed as the percentage change in quantity supplied divided by the percentage change in price.

The price elasticity of supply measures how much the quantity supplied responds to changes in the price. The supply of a good is said to be elastic if the quantity supplied responds substantially to changes in the price. Supply is said to be inelastic if the quantity supplied responds only slightly to changes in the price.

Determinants of The Price Elasticity of Supply

In most markets, a key determinant of the price elasticity of supply is ***the time period*** being considered. Supply is usually more elastic in the long run than in the short run.

Over short periods of time, firms cannot easily change the size of their factories to make more or less of a good. Thus, in the short run, the quantity supplied is not very responsive to the price.

By contrast, firms can build new factories over longer periods or close old ones. In addition, new firms can enter a market, and old firms can exit. Thus, the quantity supplied can respond substantially to price changes in the long run.

Computing the Price Elasticity of Supply

Economists compute the price elasticity of supply as the percentage change in the quantity supplied divided by the percentage change in the price. That is,

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}.$$

Computing the Price Elasticity of Supply

For example, suppose that an increase in the price of milk from \$2.85 to \$3.15 a gallon raises the amount that dairy farmers produce from 9,000 to 11,000 gallons per month. Using the midpoint method, we calculate the percentage change in price as

$$\text{Percentage change in price} = (3.15 - 2.85)/3.00 \times 100 = 10 \text{ percent.}$$

Similarly, we calculate the percentage change in quantity supplied as

$$\text{Percentage change in quantity supplied} = (11{,}000 - 9{,}000)/10{,}000 \times 100 = 20 \text{ percent.}$$

In this case, the price elasticity of supply is

$$\text{Price elasticity of supply} = \frac{20 \text{ percent}}{10 \text{ percent}} = 2.$$

In this example, the elasticity of 2 indicates that the quantity supplied changes proportionately twice as much as the price.

The Variety of Supply Curves

Economists classify demand curves according to their elasticity.

Supply is considered elastic when the elasticity is greater than 1, which means the quantity moves proportionately more than the price.

Supply is considered inelastic when the elasticity is less than 1, which means the quantity moves proportionately less than the price.

If the elasticity is exactly 1, the percentage change in quantity equals the percentage change in price, and supply is said to have unit elasticity.

The Variety of Supply Curves

a) Perfectly Inelastic Supply: Elasticity Equals 0
b) Inelastic Supply: Elasticity Is Less Than 1
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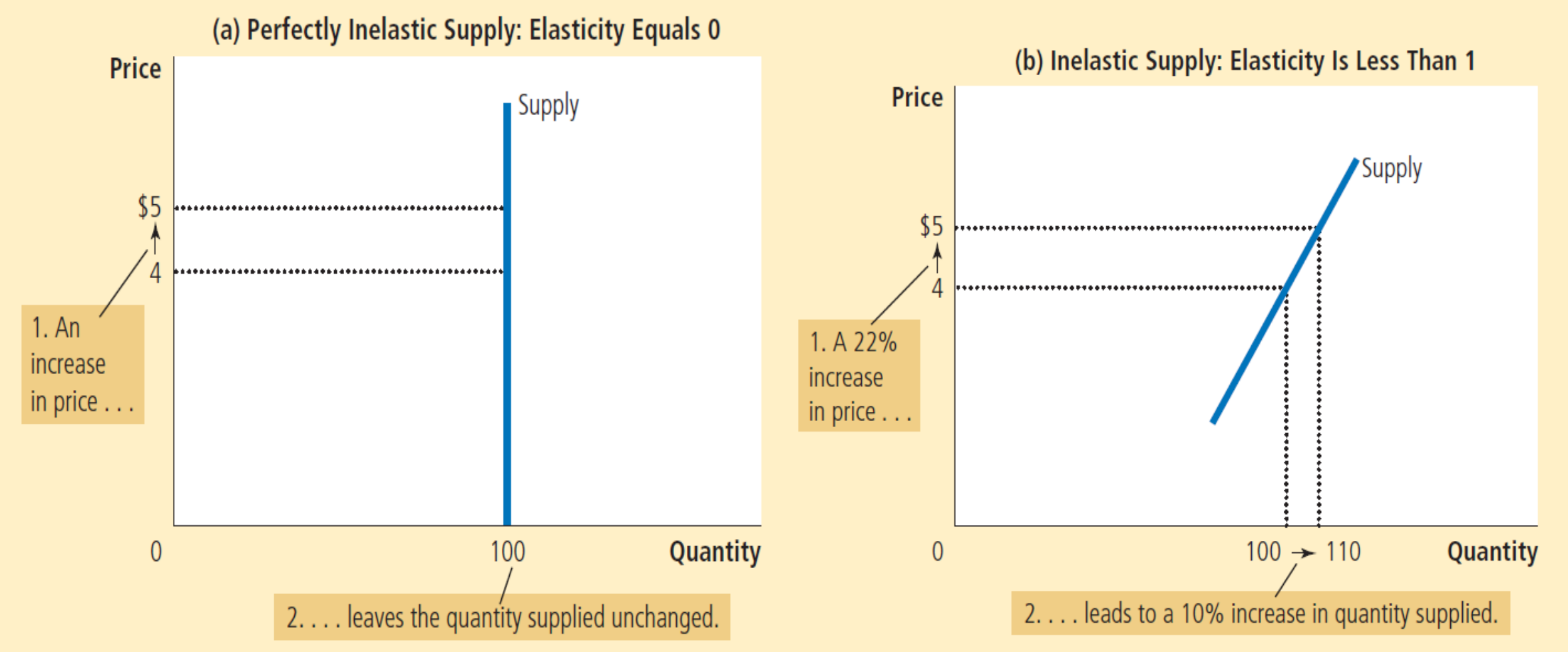
The Variety of Supply Curves



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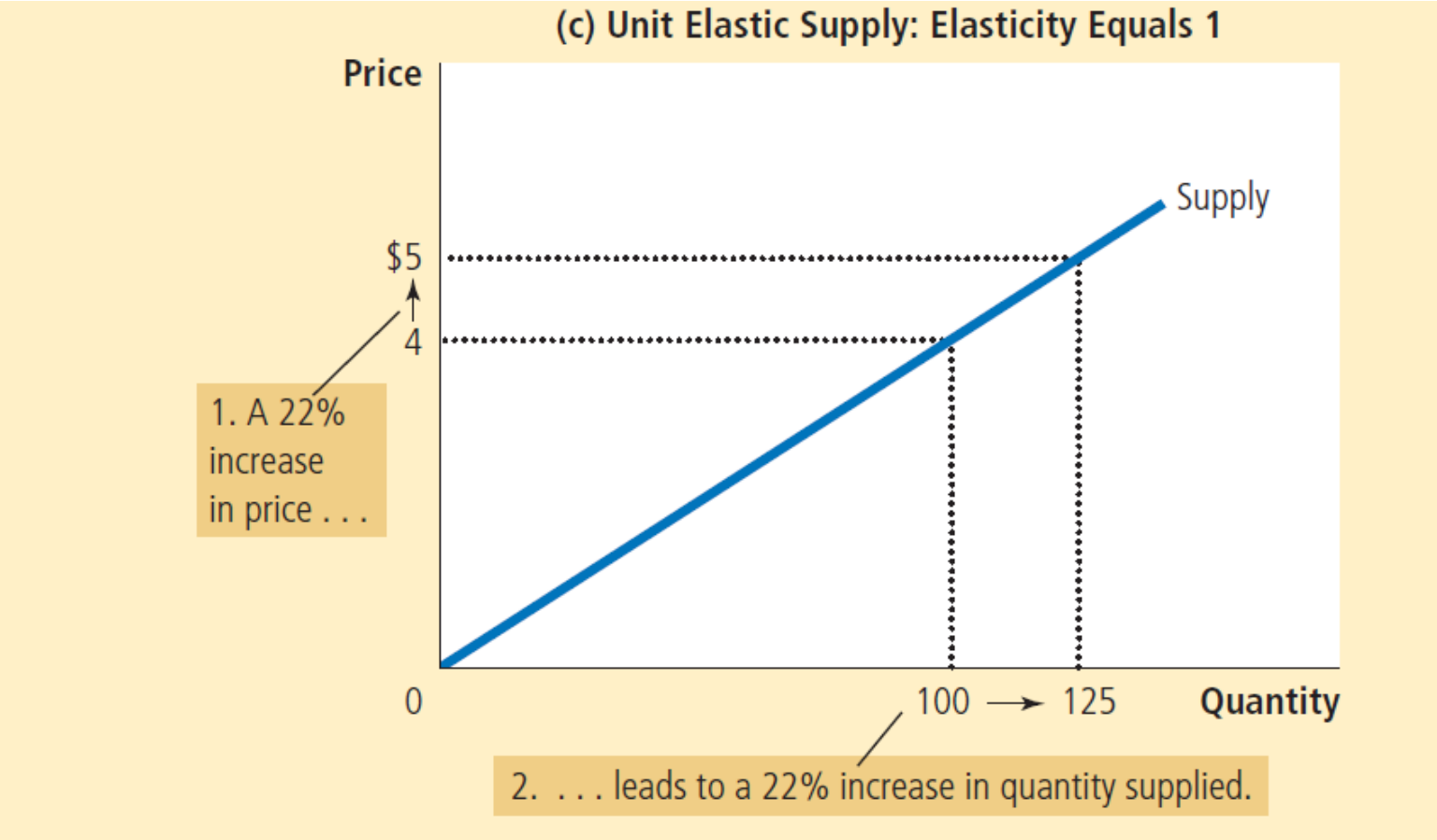
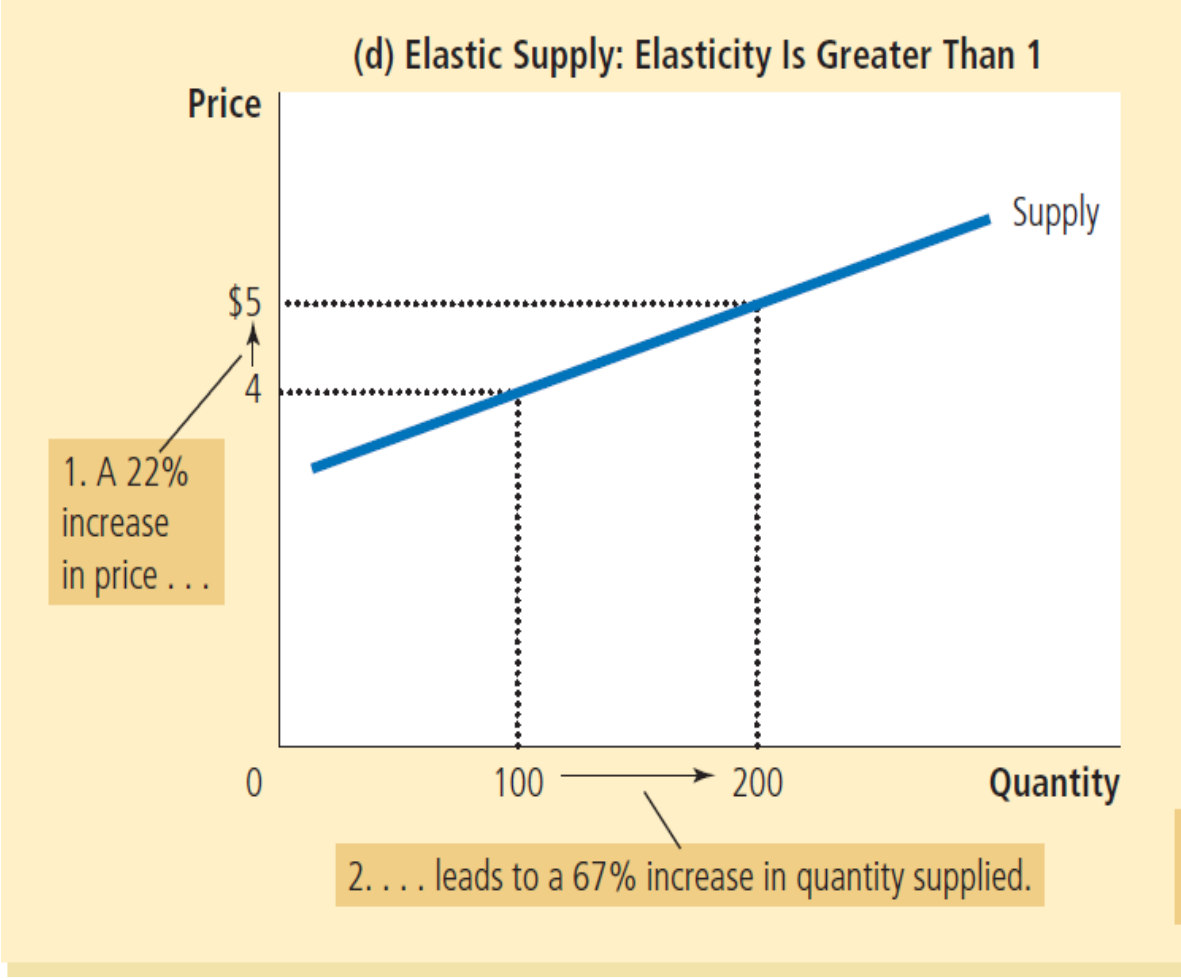
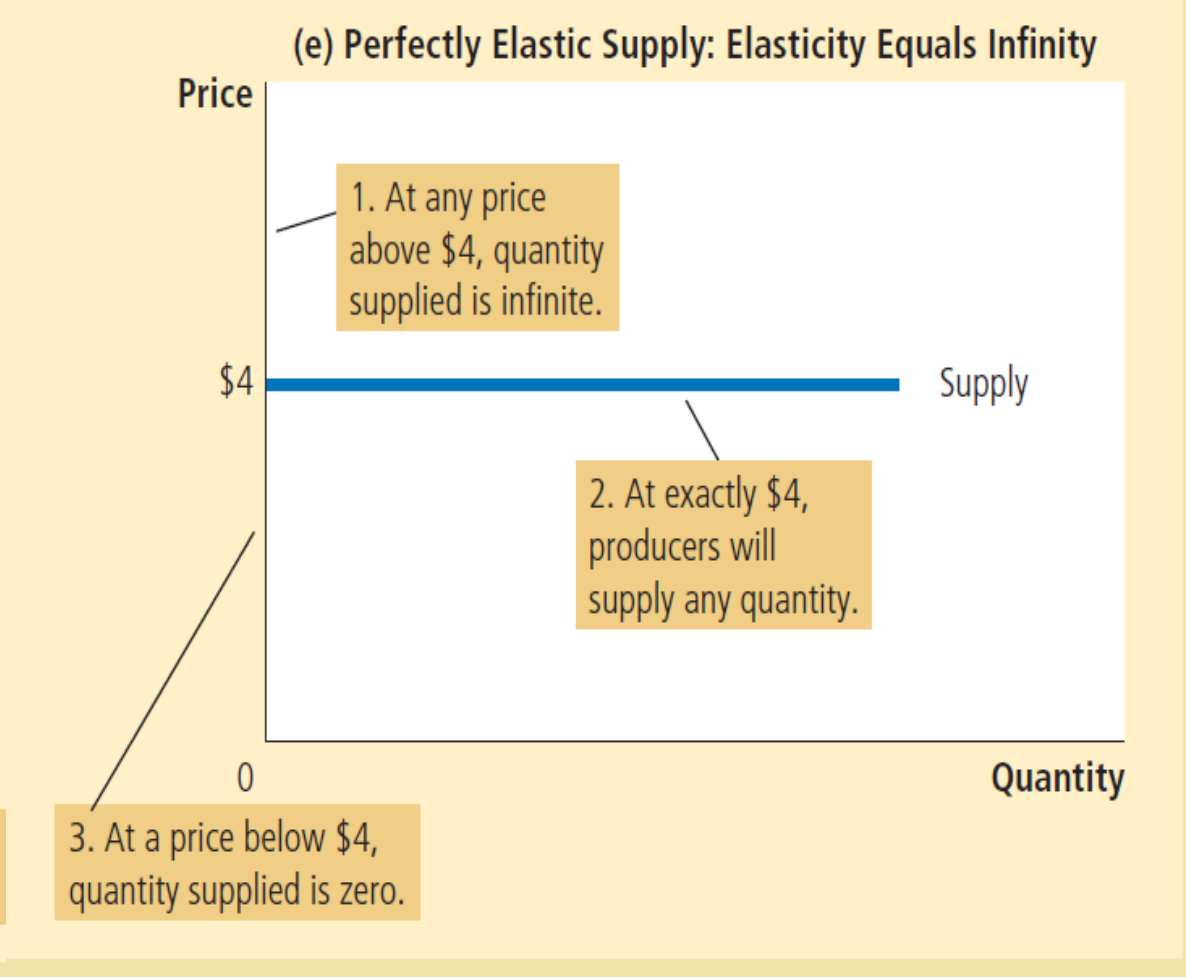


The Variety of Supply Curves





Microeconomics
Course Code: Hum 2101

Lecture 6:Market Structure

Market

Market generally means a place or a geographical area, where buyers with money and sellers with their goods meet to exchange goods for money.

In economics, the Market means a social system through which the sellers and buyers of a commodity or a service (or a group of commodities and services) can interact with each other.

Characteristics of a Market

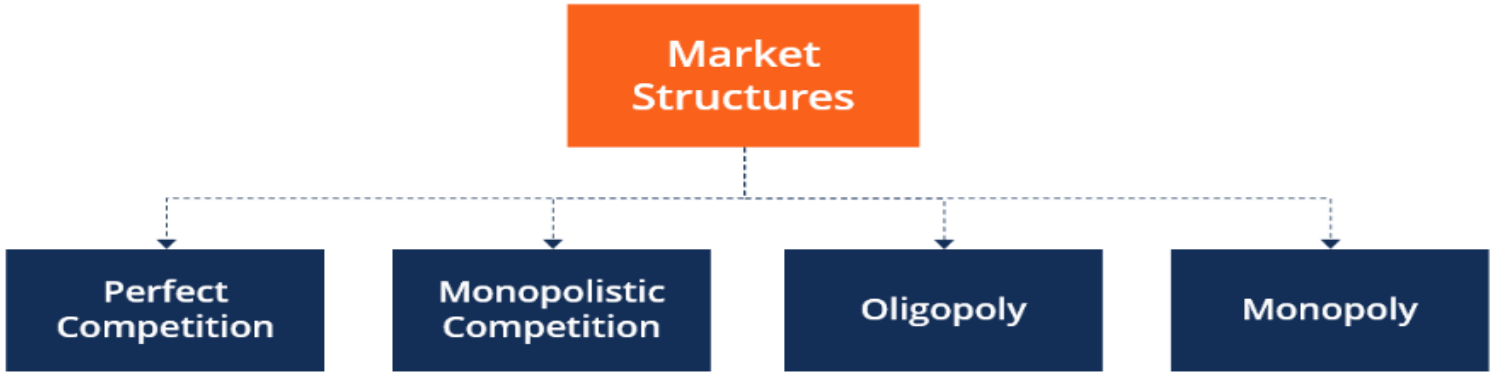
- Existence of buyers and sellers of the commodity.
- The establishment of contact between the buyers and sellers. Distance is of no consideration if buyers and sellers could contact each other through the available communication system like telephone, agents, letter correspondence, and the Internet.
- Buyers and sellers deal with the same commodity or variety. Since the market in economics is identified on the basis of the commodity, the similarity of the product is very essential.
- There should be a price for the commodity bought and sold in the market.
- The market differs from one another due to differences in the number of buyers and sellers, nature of the product, influence over price, availability of information, conditions of supply, etc.

Determinants of market structure

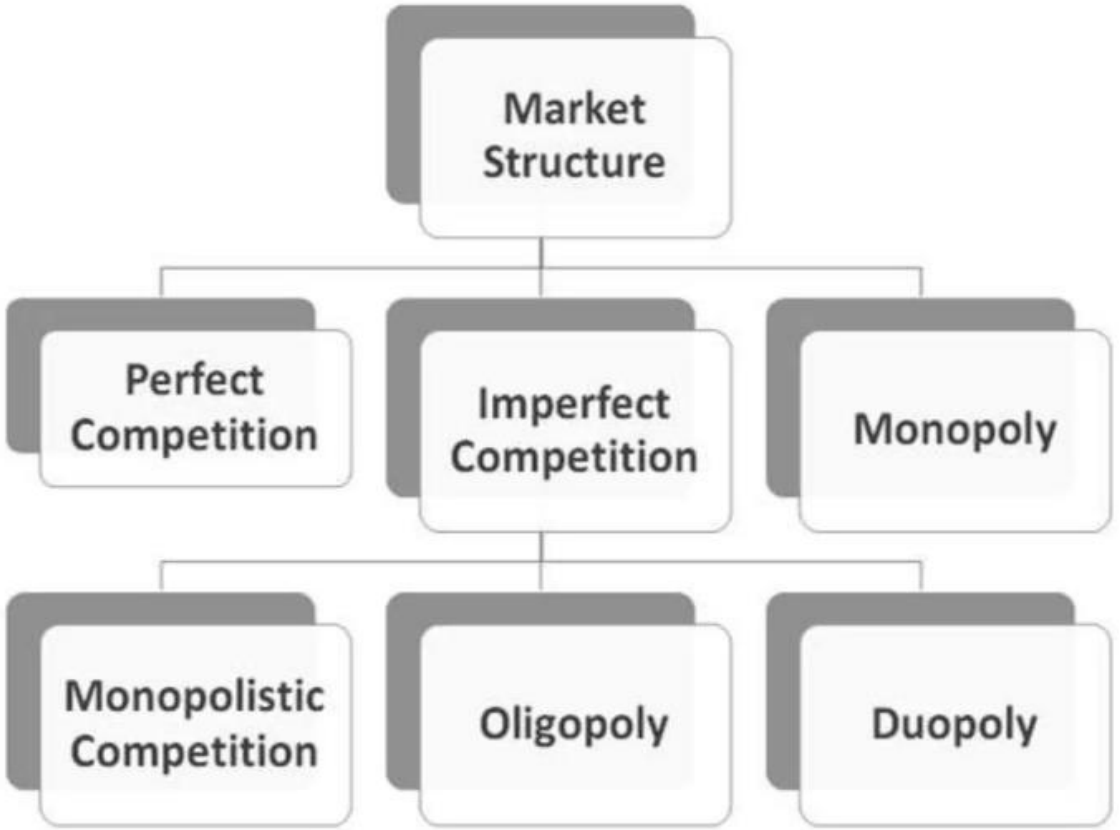
- Freedom of entry and exit
- Nature of the product
- Control over supply/output
- Control over price
- Barriers to entry

Economists discuss four broad categories of market structure, they are:

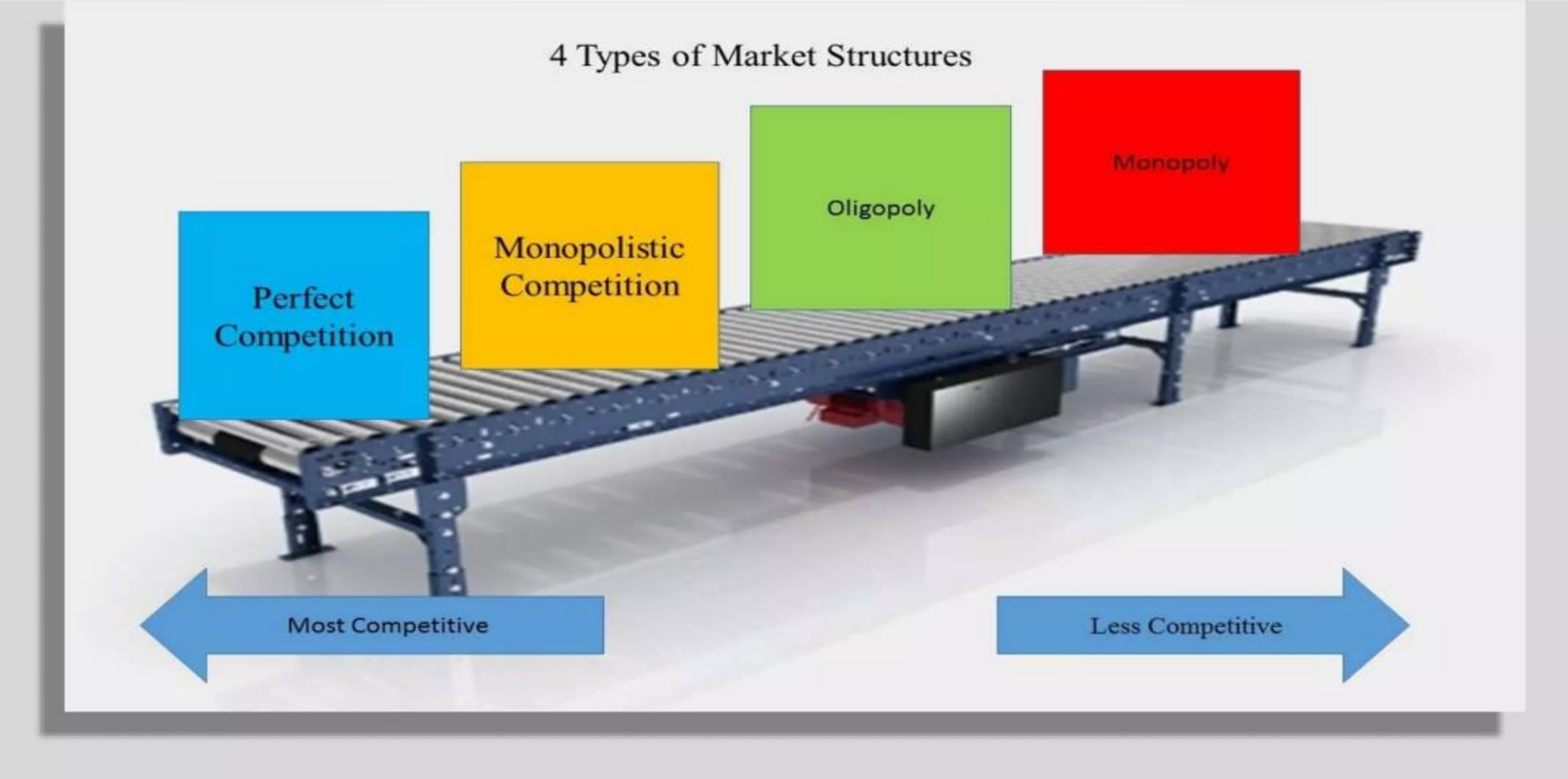
- Perfect Competition
- Monopoly
- Monopolistic Competition
- Oligopoly



Market Structure



Market Structure



Perfect Competition

Perfect competition is a market situation where there are infinite numbers of sellers that no one is big enough to have any appreciable influence over market price.

Perfect Competition

Characteristics of Perfect Competition:

1) Large number of buyers and sellers:

There are a large number of buyers and sellers in a perfectly competitive market neither a single buyer nor a single seller can influence the price.

The price is determined by market forces namely the demand for and the supply of the product. There will be uniform prices in the market. Sellers accept this price and adjust the quantity produced to maximize their profit.

Thus, sellers in the perfectly competitive market are price-takers and quantity adjusters.

Perfect Competition

2) Homogeneous Product:

The products produced by all the firms in the perfectly competitive market must be homogeneous and identical in all respects i.e. the products in the market are the same in quantity, size, taste, etc. The products of different firms are perfect substitutes.

3) Perfect knowledge of market conditions:

Both buyers and sellers are fully aware of the current price in the market. Therefore the sellers will not offer a high price and the buyers will not accept a price less than the one prevailing in the market.

Perfect Competition

4) Free Entry and Free Exit:

There must be complete freedom for the entry of new firms or the exit of the existing firms from the industry. When the existing firms are earning super-normal profits, new firms enter the market. When there is a loss in the industry, some firms leave the industry.

5) Perfect mobility of factors of production :

The factors of production should be free to move from one use to another or from one industry to another easily to get better remuneration.

The assumption of perfect mobility of factors is essential to fulfill the first condition namely a large number of producers in the market.

Perfect Competition

6) Absence of transport cost:

In a perfectly competitive market, it is assumed that there are no transport costs. Under perfect competition, a commodity is sold at a uniform price throughout the market. If transport cost is incurred, the firms nearer to the market will charge a low price than the firms far away. Hence it is assumed that there is no transport cost.

7) Absence of Government or artificial restrictions or collisions:

There are no government controls or restrictions on supply, pricing, etc. There is also no collusion among buyers or sellers. The price in a perfectly competitive market is free to change in response to changes in demand and supply conditions

Perfect Competition

Advantages of Perfect Competition-

1. It discourages customer exploitation
2. It creates an active business environment
3. There are no restrictions to enter or exit the market
4. Customers get high-quality products at reasonable price
5. Producers focus on maximum utilization of resources to improve efficiency
6. It requires less advertising and publicity budget

Perfect Competition

Disadvantages of Perfect Competition-

1. It is difficult for small traders to sustain themselves because of high competition
2. It lacks product differentiation
3. Firms cannot enjoy the benefit of economies of scale
4. In this market, the profit margin is very low. So producers cannot make sufficient investments for research and development purposes.
5. Customers get a limited choice

Monopoly

Monopoly is a market structure in which there is a single seller, there are no close substitutes for the commodity it produces and there are barriers to entry.

Monopoly

Characteristics of Monopoly:

1. Single Seller:

There is only one seller; he can control either price or supply of his product. But he cannot control demand for the product, as there are many buyers.

2. No close Substitutes:

There are no close substitutes for the product. The buyers have no alternatives or choices. Either they have to buy the product or go without it.

Monopoly

3. Price:

The monopolist has control over the supply so as to increase the price. Sometimes he may adopt price discrimination. He may fix different prices for different sets of consumers. He is a price-maker, not a price-taker.

4. No Entry:

There is no freedom for other producers to enter the market as the monopolist is enjoying monopoly power. There are strong barriers for new firms to enter. There are legal, technological, economic, and natural obstacles, which may block the entry of new producers. He uses patent rights.

5. Firm and Industry:

Under a monopoly, there is no difference between a firm and an industry. As there is only one firm, that single firm constitutes the whole industry.

Monopoly

Advantages of Monopoly:

1. Production: Monopoly firms have large-scale production possibilities. This will result in the reduction of costs of production. Output can be sold at low prices. This is beneficial to the consumers.
2. Resources: Monopoly firms have vast financial resources which could be used for research & development.
3. Merger: There are a number of weak firms in an industry. These firms can combine together in the form of a monopoly to meet competition. In such a case, the market can be expanded.

Monopoly

Disadvantages of Monopoly

1. High Price: A monopolist always charges a high price, which is higher than the competitive price. Thus a monopolist exploits the consumers.
2. Maximum Profit: A monopolist is interested in getting maximum profit. He may restrict the output and raise prices. Thus, he creates artificial scarcity for his product.
3. Price Differentiation: A monopolist often charges different prices for the same product from different consumers.
4. Self-interest: A monopolist uses large-scale production and huge resources to promote his own selfish interest. He may adopt wrong practices to establish absolute monopoly power.
5. Not following the Socialist or Mixed Economy Concept: In a country dominated by monopolies, wealth is concentrated in the hands of a few. It will lead to inequality of income.

Monopolistic Competition

Monopolistic competition, as the name itself implies, is a blending of monopoly and competition. Monopolistic competition refers to the market situation in which a large number of sellers produce goods that are close substitutes for one another. The products are similar but not identical.

The particular brand of the product will have a group of loyal consumers. In this respect, each firm will have some monopoly, and at the same time, the firm has to compete in the market with the other firms as they produce a fair substitute.

The following are examples of monopolistic competition in the Bangladesh context.

I. Shampoo - Sun Silk, Clinic Plus, Ponds, Head and Shoulder, Pantene, Vatika, etc.

II. Tooth Paste - Colgate, Close-up, Promise, Pepsodent, etc.

Monopolistic Competition

Features of Monopolistic Competition:

1. Large number of sellers and buyers
2. Product differentiation – Closely related products
3. Non-Price Competition – Spends large amounts of money on advertisements keeping the product prices unchanged.
4. Free entry and Free exit
5. Independent price policy

Oligopoly

Oligopoly refers to a form of imperfect competition where there will be only a few sellers producing either homogenous or differentiated products.

The numbers of firms are so small that every seller is affected by the activities of the others.

Oligopoly

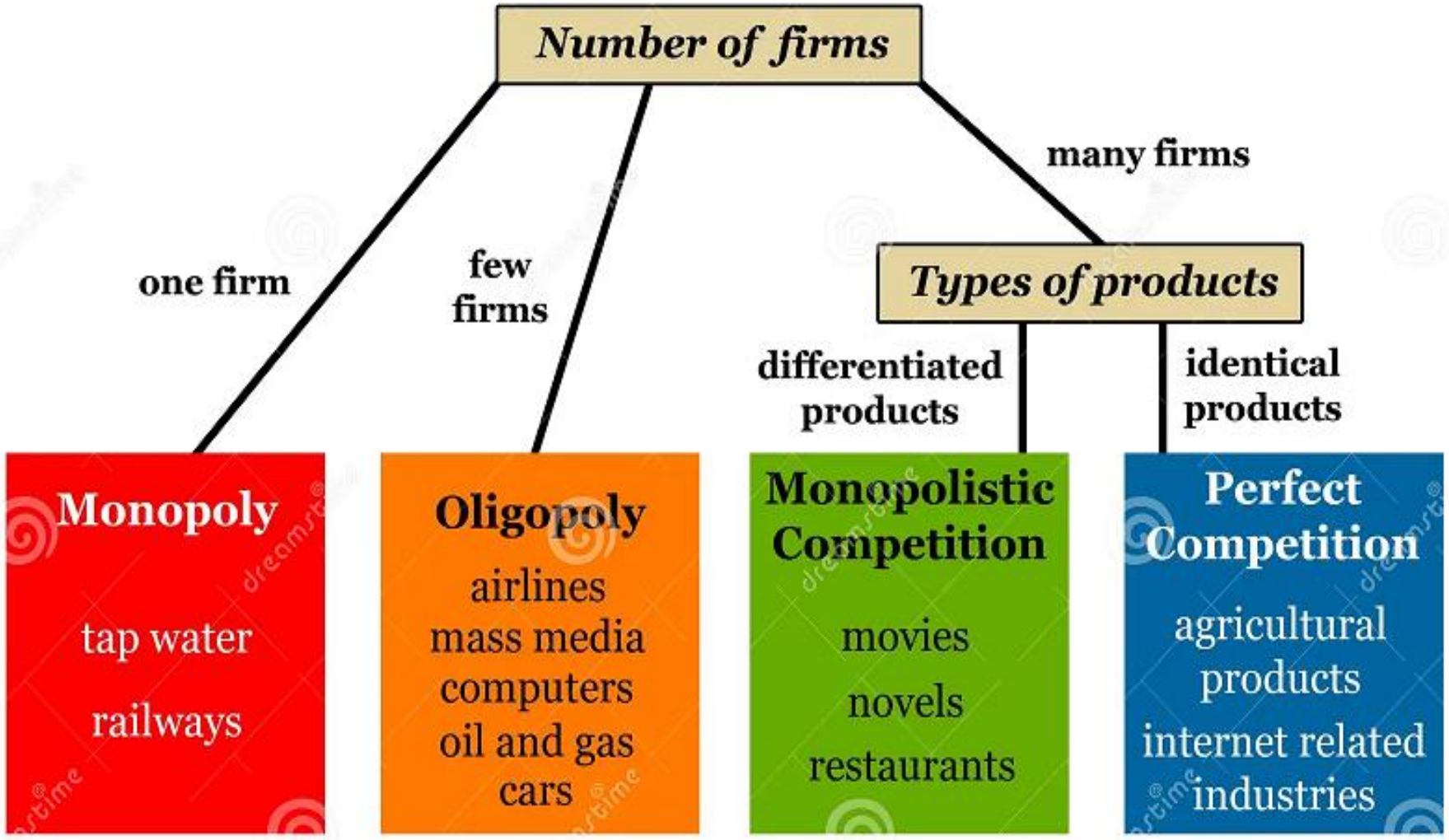
Characteristics of Oligopoly:

- Few Sellers
- Control over supply
- Inter-Dependence of firms
- Conflicting attitudes of firms
- Advertising and selling cost
- Price rigidity
- Group behavior
- Intense competition

Market Structures

Features	Perfect Competition	Monopolistic Competition	Oligopoly	Monopoly
No Of Producers (Sellers)	Large No Of Producers	Large No Of Producers	Few Large Producers	Only One Producer
Product Features	Identical / Non-Differentiated	Highly Similar Highly Substitutable But Not Identical	Differentiated	Unique No Substitutes
Industry Entry	Easy Industry Entry & Exit	Easy Industry Entry & Exit	High Barriers to Entry But Not Impossible	Very High Barriers to Entry Nearly Impossible To Entry/Exit
Competition	High	High	Competitors Act In A Collaborative Manner (Mutual Interdependent)	No Competition / Low
Market Power of Individual Firm (Ability to Control the Price)	None	Has Low Control Over Price	High Market Power Can Influence The Price With Collaborative Manner	Very High Has the Complete Control Over Price Control

The four types of Market Structure



Microeconomics
Course Code: Hum 2101

Lecture 7: Consumer Behavior Theory

Consumption Choices

The choices that you make as a buyer of goods and services—your consumption choices—are influenced by many factors. We can summarize them into two broad headings:

- Consumption possibilities
- Preferences

Consumption Possibilities

Your consumption possibilities are all the things that you can afford to buy. You can afford many different combinations of goods and services, but they are all limited by your income and by the prices that you must pay.

For example, you might decide to spend a big part of your income on a gym membership and personal trainer and little on movies and music, or you might spend lots on movies and music and use the free gym at school.

The easiest way to describe consumption possibilities is to consider a model consumer who buys only two items. That’s what we’ll now do. We’ll study the consumption possibilities of Lisa, who buys only movies and soda.

Consumption Possibilities

A Consumer's Budget Line

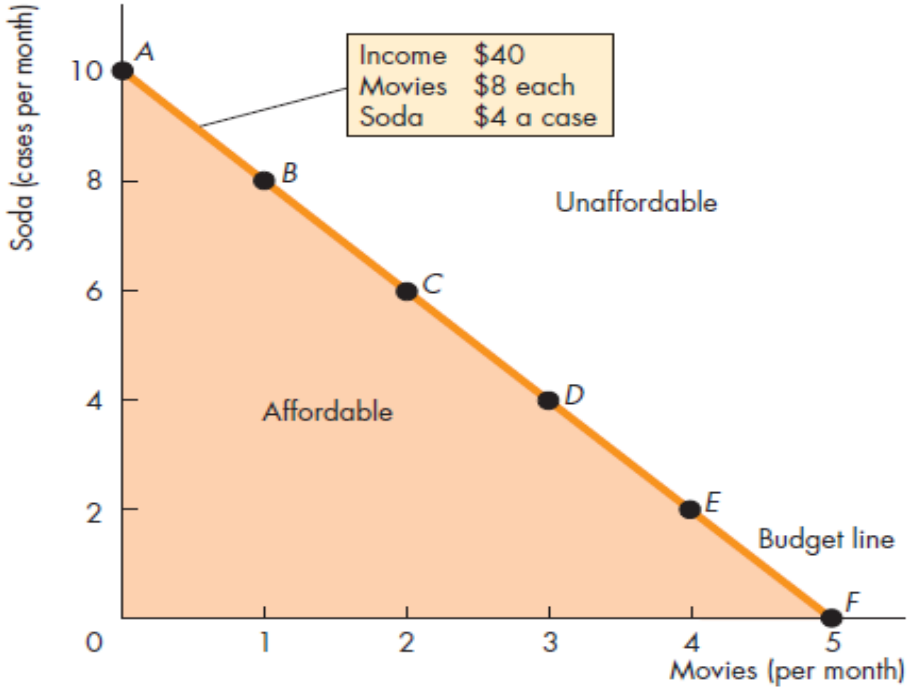
Consumption possibilities are limited by income and by the prices of movies and soda. When Lisa spends all her income, she reaches the limits of her consumption possibilities. We describe this limit with a ***budget line***, which marks the boundary between those combinations of goods and services that a household can afford to buy and those that it cannot afford.

Consumption Possibilities

Possibility	Movies Quantity	Movies Expenditure (dollars)	Soda Cases	Soda Expenditure (dollars)
A	0	0	10	40
B	1	8	8	32
C	**2**	**16**	**6**	**24**
D	3	24	4	16
E	4	32	2	8
F	5	40	0	0

The graph and the table show six possible ways in which Lisa can allocate $40 to movies and soda. In row *C* and at point *C*, she sees 2 movies and buys 6 cases of soda. The line *AF* is Lisa's budget line and is a boundary between what she can afford and what she cannot afford. Her choices must lie along the line *AF* or inside the orange area.

FIGURE 8.1 Lisa's Budget Line



Consumption Possibilities

This figure shows Lisa's consumption possibilities of movies and soda and her budget line.

Lisa has an income of $40 a month, the price of a movie is $8, and the price of soda is $4 a case.

Rows A through F in the table show six possible ways of allocating $40 to these two goods.

For example, in row A Lisa buys 10 cases of soda and sees no movies; in row F she sees 5 movies and buys no soda; in row C she sees 2 movies and buys 6 cases of soda. Points A through F in the graph illustrate the possibilities presented in the table, and the line passing through these points is Lisa's budget line.

The budget line constrains choices: It marks the boundary between what is affordable and unaffordable. Lisa can afford all the points on the budget line and inside it. Points outside the line are unaffordable.

Consumption Possibilities

Changes in Consumption Possibilities

Consumption possibilities change when ***income or prices change***.

A rise in income shifts the budget line outward. Our goal is to predict the effects of such changes on consumption choices. To do so, we must determine the choice a consumer makes.

The budget line shows what is possible; preferences determine which possibility is chosen. We'll now describe a consumer's preferences.

Preferences

Lisa's income and the prices that she faces limit her consumption choices, but she still has lots of choices. The choice that she makes depends on her preferences—a description of her likes and dislikes.

The goal of a theory of consumer choice is to derive the demand curve from a deeper account of how consumers make their buying plans. That is, we want to explain what determines demand and marginal benefit.

To achieve this goal, we need a deeper way of describing preferences. One approach to this problem uses the idea of utility, and *defines utility as the benefit or satisfaction that a person gets from the consumption of goods and services.* We distinguish two utility concepts:

- Total utility
- Marginal utility

Preferences

Total Utility

The total benefit that a person gets from the consumption of all the different goods and services is called total utility.

Total utility depends on the level of consumption—more consumption generally gives more total utility.

To illustrate the concept of total utility, think about Lisa's choices.

We give her two starting points:

(1) We will call the total utility from no movies and no soda zero utility;

(2) We will call the total utility she gets from seeing 1 movie a month 50 units.

We then asked Lisa to tell us, using the same scale, how much she would like 2 movies, and more, up to 10 movies a month. We also asked her to tell us, on the same scale, how much she would like 1 case of soda a month, 2 cases, and more, up to 10 cases a month.

Preferences

Marginal Utility

The expression “marginal” is a key term in economics and always means “additional” or “extra.” Marginal utility denotes the additional utility you get from the consumption of an additional unit of a commodity.

One of the fundamental ideas behind demand theory is the law of diminishing marginal utility. This law states that the amount of extra or marginal utility declines as a person consumes more and more of a good. To understand this law, first, remember that utility tends to increase as you consume more of a good. However, as you consume more and more, your total utility will grow at a slower and slower rate. This is the same thing as saying that your marginal utility (the extra utility added by the last unit consumed of a good) diminishes as more of a good is consumed.

The law of diminishing marginal utility states that, as the amount of a good consumed increases, the marginal utility of that good tends to decline.

Movies			Soda		
Quantity (per month)	Total utility	Marginal utility	Cases (per month)	Total utility	Marginal utility
0	0	 50	0	0	 75
1	50	 40	1	75	 48
2	90	 32	2	123	 36
3	122	 28	3	159	 24
4	150	 26	4	183	 22
5	176	 24	5	205	 20
6	200	 22	6	225	 13
7	222	 20	7	238	 10
8	242	 17	8	248	 7
9	259	 16	9	255	 5
10	275		10	260	

Preferences

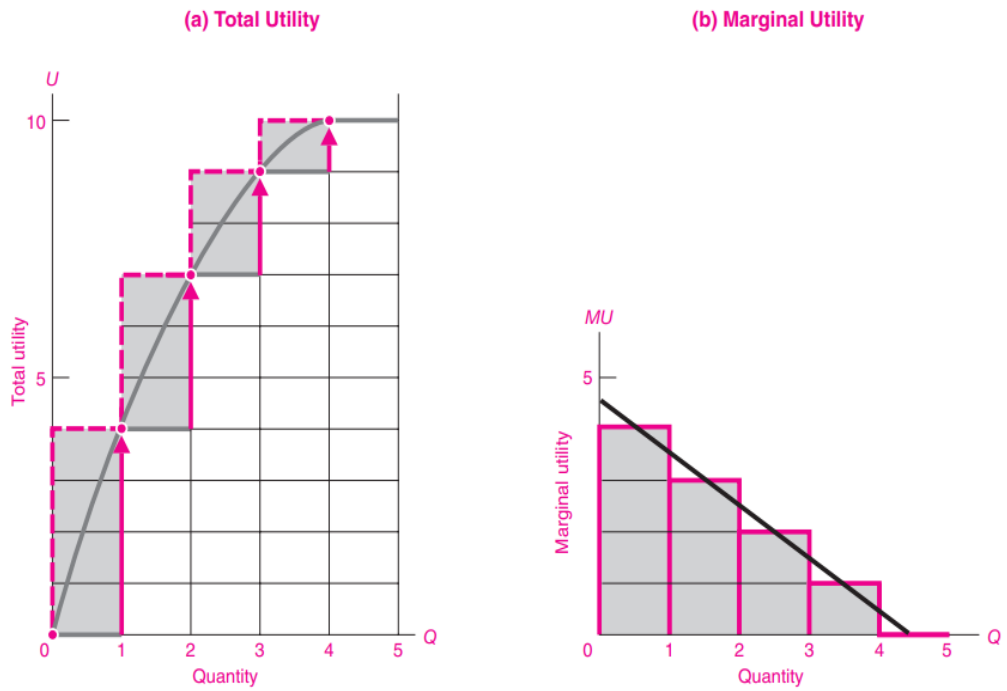
Total Utility and Marginal Utility

- Looking at these numbers, you can say a lot about how much Lisa likes soda and movies.
- She says that 1 case of soda gives her 75 units of utility—50 percent more than the utility that she gets from seeing 1 movie.
- You can also see that her total utility from soda climbs more slowly than her total utility from movies.

- "Marginal utility" column shows Lisa's marginal utility from movies and soda. You can see that if Lisa increases the soda she buys from 1 to 2 cases a month, her total utility from soda increases from 75 units to 123 units.
- For Lisa, the marginal utility from the second case each month is 48 units (123 – 75).
- The marginal utility numbers appear midway between the quantities of soda because it is the change in the quantity she buys from 1 to 2 cases that produces the marginal utility of 48 units.
- Marginal utility is positive, but it diminishes as the quantity of a good consumed increases.

Preferences

Total Utility and Marginal Utility



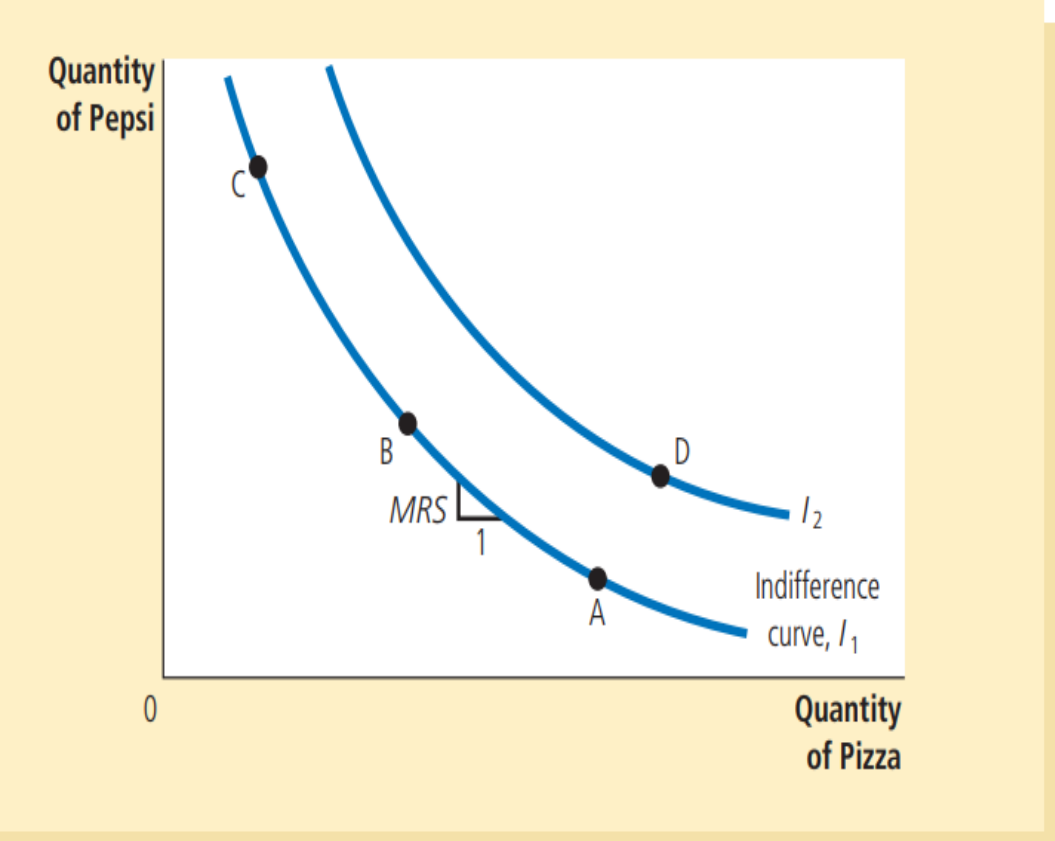
- Total utility in (a) rises with consumption, but it rises at a decreasing rate, showing diminishing marginal utility. This observation led early economists to formulate the law of downward sloping demand.
- The blue blocks show the extra utility added by each new unit. The fact that total utility increases at a decreasing rate is shown in (b) by the declining steps of marginal utility. If we make our units smaller, the steps in total utility are smoothed out and total utility becomes the smooth blue curve in (a). Moreover, smoothed marginal utility, shown in (b) by the blue downward-sloping smooth curve, becomes indistinguishable from the slope of the smooth curve in (a

Representing Preferences with Indifference Curves

Indifference curve - A curve that shows consumption bundles that give the consumer the same level of satisfaction.

The consumer's preferences allow her to choose among different bundles of pizza and Pepsi. If you offer the consumer two different bundles, she chooses the bundle that best suits her tastes. If the two bundles suit her tastes equally well, we say that the consumer is indifferent between the two bundles.

Representing Preferences with Indifference Curves



- This figure shows two of the consumer's many indifference curves.
- The consumer is indifferent among combinations A, B, and C because they are all on the same curve.
- Not surprisingly, if the consumer's consumption of pizza is reduced, say, from point A to point B, consumption of Pepsi must increase to keep her equally happy.
- If consumption of pizza is reduced again, from point B to point C, the amount of Pepsi consumed must increase yet again.

Representing Preferences with Indifference Curves

Marginal rate of substitution - the rate at which a consumer is willing to trade one good for another.

The slope at any point on an indifference curve equals the rate at which the consumer is willing to substitute one good for the other. This rate is called the marginal rate of substitution (MRS).

In this case, the marginal rate of substitution measures how much Pepsi the consumer requires to be compensated for a one-unit reduction in pizza consumption. Notice that because the indifference curves are not straight lines, the marginal rate of substitution is not the same at all points on a given indifference curve.

The rate at which a consumer is willing to trade one good for the other depends on the amount of the goods she is already consuming. That is, the rate at which a consumer is willing to trade pizza for Pepsi depends on whether she is hungrier or thirstier, which in turn depends on how much pizza and Pepsi she is consuming

Representing Preferences with Indifference Curves

Four Properties of Indifference Curves

- Property 1: Higher indifference curves are preferred to lower ones.
- Property 2: Indifference curves are downward sloping.
- Property 3: Indifference curves do not cross.
- Property 4: Indifference curves are bowed inward.